

INTRODUCTION

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- Education data provides valuable insight into academic outcomes and school operations.
- This project explores student performance, grade levels, staffing, and salary distribution across schools.
- The goal is to uncover patterns in school success and performance using SQL, Python, and Power BI.
- Insights are derived from a normalized Oracle database structured around real-world academic entities.

RESEARCH QUESTIONS¹

- What factors influence overall student performance in different schools?
- How do grade levels correlate with academic success and resources?
- What trends exist in teacher and principal salaries across schools?
- How can data visualization assist in strategic educational planning?

DATA MODELING PROCESS

- **Data Source:** An Oracle SQL schema with 8 interconnected academic tables.
- **Tools Used:** Oracle SQL Developer, Python (pandas), and Power BI.
- **Process:**
 - Extracted and cleaned data using SQL and Python.
 - Modeled relationships across schools, staff, students, and grades.
 - Created visualizations to explore patterns in performance and finance.
- **Analysis Techniques:**
 - Aggregations, joins, and filters to isolate insights.
 - DAX measures to calculate ratios and averages.
 - Python used for data cleaning and CSV generation.

TOTAL SCHOOLS

3

TOTAL TEACHERS

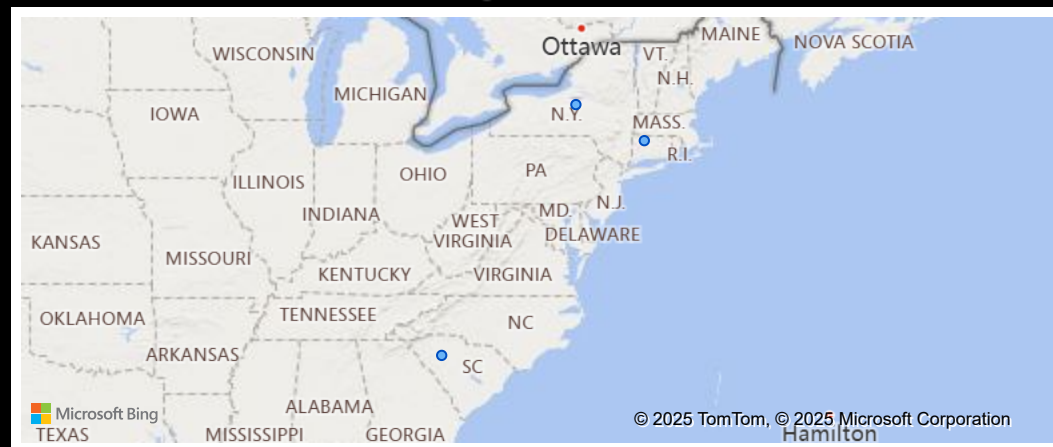
15

TOTAL STUDENTS

72

SCHOOL OVERVIEW

MAP OR TABLE

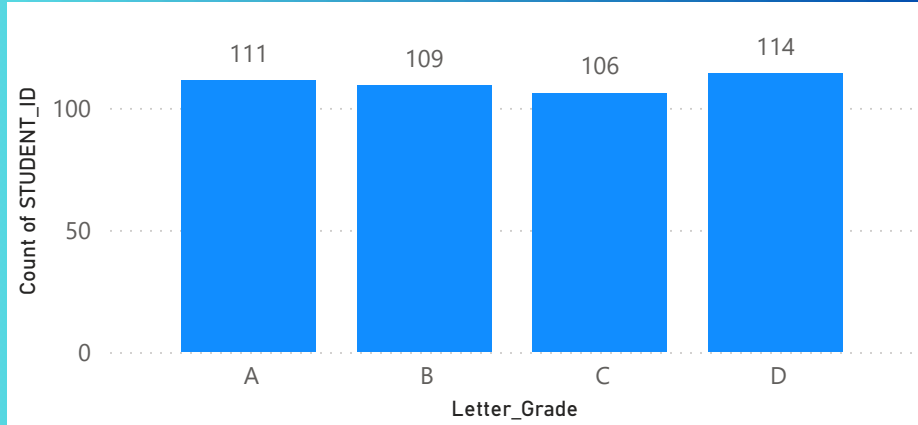


AVERAGE STUDENTS PER SCHOOL

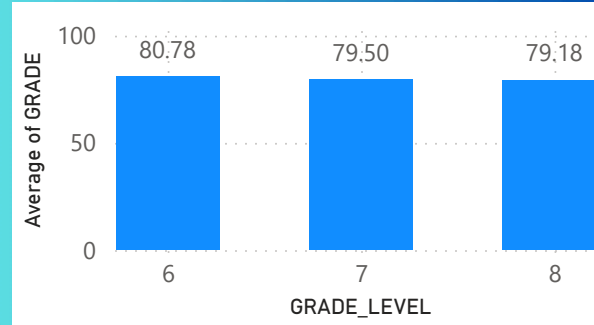
24.00

SCHOOL ANALYSIS

OVERALL GRADE DISTRIBUTION



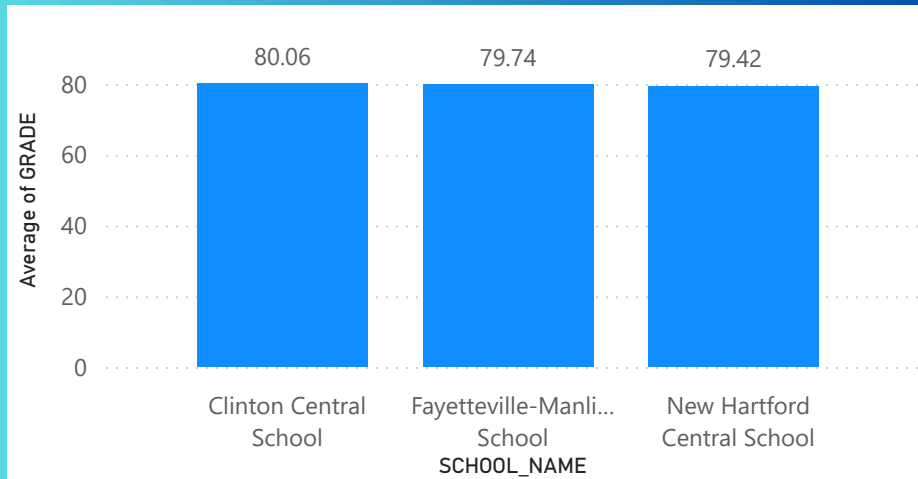
TOP PERFORMING GRADE LEVELS



AVERAGE STUDENT GRADE

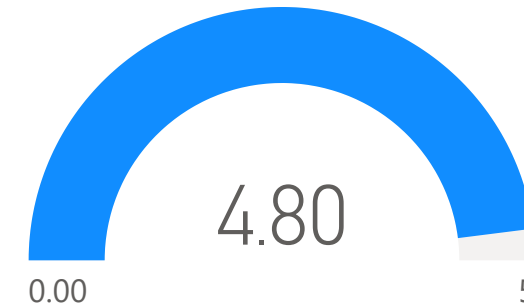
79.78

STUDENTS PERFORMANCE BY SCHOOL



STUDENT-TO-TEACHER -RATIO

4.80



FINANCIAL INSIGHTS

TOTAL PRINCIPAL
SALARIES

\$243.23K

TOTAL TEACHER
SALARIES

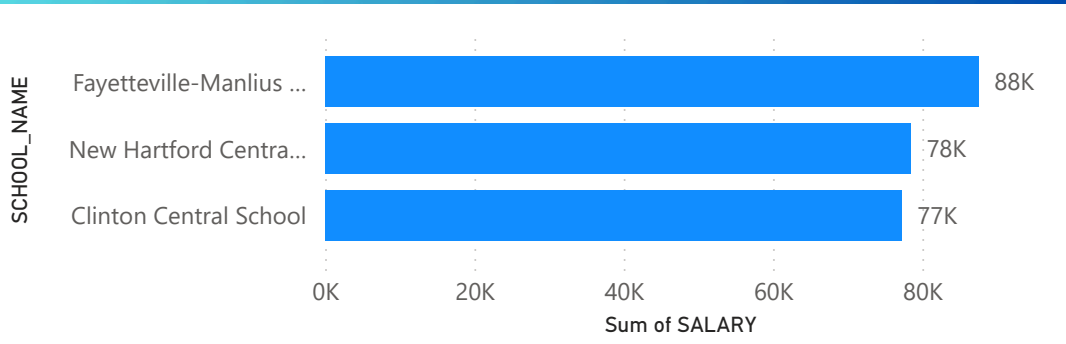
\$695.58K

AVERAGE SALARIES

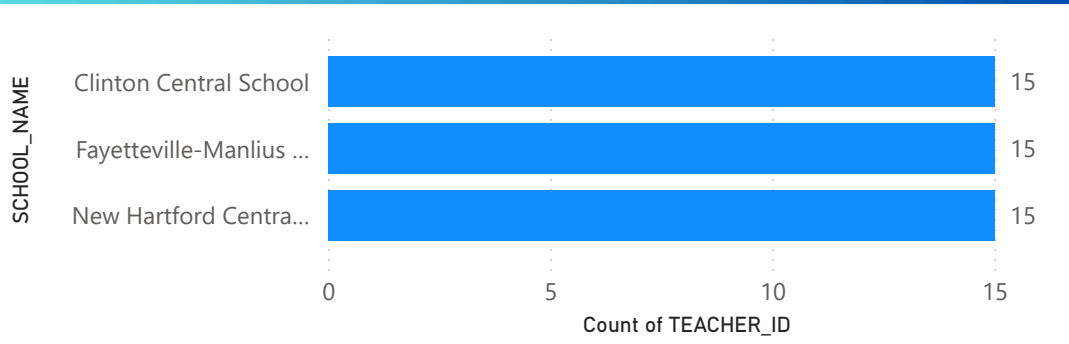
\$81.08K

\$46.37K

TOP SCHOOLS BY PRINCIPAL SALARY



TOP SCHOOLS BY NUMBER OF TEACHERS



HIGHEST PAID PRINCIPAL

\$87.54K

FIRST_NAME	LAST_NAME	SCHOOL_NAME	Sum of SALARY
Rachel	Alexander	Fayetteville-Manlius School	87537
Total			87537

HIGHEST PAID TEACHER

\$61.07K

FIRST_NAME	LAST_NAME	SCHOOL_NAME	SALARY
Abigail	Lopez	Clinton Central School	61068

BUSINESS PROBLEM:

Which school should receive a budget increase to improve academic performance and sustain staff retention?

INSIGHTS:

- Schools with lower student-to-teacher ratios are achieving higher academic performance. (Smaller class sizes allow for more individualized attention.)
- Some high-performing schools are operating with a lower average teacher salary, risking potential staff turnover and future decline in performance.

Recommendation:

Prioritize Fayetteville-Manlius School for a budget increase.

Use additional funds to increase teacher salaries and reinforce academic excellence.

CONCLUSION

- Fayetteville-Manlius shows strong academic potential
- Smaller class sizes correlate with higher performance
- Budget reallocation can reinforce outcomes
- Schools with higher-performing students also tend to have lower student-to-teacher ratios and more experienced staff, suggesting that resource allocation and staffing quality directly impact academic outcomes.
- High student performance does not always align with high staff pay—indicating a possible disconnect that may impact future teacher retention and reward systems.